



Effect of pesticides on nitrification activity and its interaction with chemical fertilizer and manure in long-term paddy soils

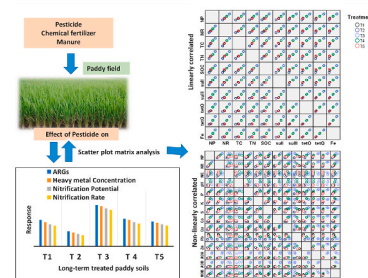
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HIGHLIGHTS

- Nitrification activities were significantly ($p < 0.05$) decreased in pesticide treated paddy soils.
- Heavy metal concentrations were significantly ($p < 0.05$) increased in chemical fertilizer with manure treated paddy soils.
- Total carbon, soil organic carbon, total nitrogen, and Fe^{2+} linearly correlated with NR and NP by Scatter plot matrix.
- ARGs and ammonia monooxygenase (*amoA*) genes were significant decrease ($p < 0.05$) in pesticide treated paddy soils.
- Effect of pesticides and its combined effect with chemical fertilizer and manure present in the paddy soils.

GRAPHICAL ABSTRACT



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ABSTRACT

Effect of pesticides on nitrification activity and its interaction among heavy metal concentrations (HMCs), antibiotic resistance genes (ARGs), and ammonia monooxygenase (*amoA*) genes of long-term paddy soils is little known. The aim was to study the effect of pesticides on net nitrification rate (NR), potential nitrification rate (NP), HMCs, ARGs (*sull*, *sulll*, *tetO*, and *tetQ*), and *amoA* (*amoA*-AOA, *amoA*-AOB, and *amoA*-NOB) genes in long-term treated paddy soils. NR and NP were significantly decreased ($p < 0.05$), whereas HMCs (Pb^{2+} , Cu^{2+} , Zn^{2+} , and Fe^{3+}) were a significantly increased ($p < 0.05$) in chemical fertilizer with pesticide treated paddy soils as compared with chemical fertilizer treated paddy soils. The scatter plot matrix indicated that total carbon (TC), soil organic carbon (SOC), total nitrogen (TN), and Fe were linearly correlated with NR and NP in long-term treated paddy soils. ARGs and *amoA* genes were significantly decreased ($p < 0.05$) in chemical fertilizer and manure with pesticide treated paddy soils. Overall, the result indicated the response of pesticide and their combination of manure with pesticide interaction present in long-term paddy soils, which will play a great role in the control uses of pesticides, manure, and chemical fertilizers in paddy soils and protect the nitrogen cycle as well as environment.

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1. Introduction

Paddy rice is measured as the main food in everyday life in many countries as well as in Bangladesh and paddy rice production substantially affects worldwide food security (Nahar et al., 2020; Rahman et al., 2020, 2021; Lu et al., 2021). In Bangladesh, paddy rice is the main concern for countrywide food safety (Mainuddin and Kirby, 2015; Ara et al., 2016; Nahar et al., 2020). Paddy rice contains high calories in nutritional quality among cereal crops (Imsong et al., 2015). It also offers carbohydrates, fiber, minerals, vitamins, etc. (Zubair et al., 2015; Pandey and Salunkhe 2017). The population of Bangladesh is increasing day by day, therefore needs to be increased much paddy production for fulfilling the demand of an increasing population (Nahar et al., 2020; Rahman et al., 2021). For this reason, farmers practice pesticides and chemical fertilizers in the paddy field for the growth of paddy rice production with inadequate land (Rahman et al., 2020; Prashar and Shah, 2016; Kakar et al., 2019). High quantity of chemical pesticides is applied to soil for pest control; therefore, these chemicals enter into the soils from above ground (Tudi et al., 2021; De Ryck et al., 2021) and effect on sensitive microorganisms (Li et al., 2021; Kakar et al., 2019).

The use of pesticide and chemical fertilizer is essential for paddy production, but overuse of them had also affected nitrification rate and nitrifying microorganisms in paddy soil by accumulating heavy metals (Pahalvi et al., 2021; Ali et al., 2021; Prashar and Shah, 2016). Previous studies reported that high levels of potassium and sodium comprising fertilizers had a negative effect on the soil structure and pH (Roberts, 2014; Diacono and Montemurro, 2015). It was also reported that the constant practice of nitrogen fertilizers creates a decline of soil liming and pH (Ashitha et al., 2021; Zhou et al., 2014; Daba et al., 2021; Ning et al., 2020; Jin et al., 2019). By using massive quantities of K^+ , Fe^{3+} , Zn^{2+} , and Ca^{2+} disturb the balance of plant nutrients and affect the organisms negatively such as the lethal effect of worms and soil mites (Ashitha et al., 2021; Amjadian et al., 2021; Shahid et al., 2016). Chemical fertilizers, manure, and pesticides have also an adverse effect on antibiotic resistance genes (Wang et al., 2014; Rahman et al., 2018). Accumulation of heavy metals transferred into the food chain in paddy soil and caused the possible health hazard (Mao et al., 2019; Ali et al., 2019; Lian et al., 2019; Kumar et al., 2019; Khan et al., 2015). Recently, it has been observed that heavy metals reduce microbial biomass in paddy soil (Deng et al., 2015; Bashir et al., 2018). It was mentioned that heavy metal affects the microbial-mediated biogeochemical methods (Bai et al., 2020; Zhao et al., 2019; Rajkumar et al., 2012). Heavy metal also suppressed soil nitrification rates both in spoil soil samples and in polluted fields (Gorovtsov et al., 2019; Klimek et al., 2021; Nahar et al., 2020; Stefanowicz et al., 2020).

Nitrification is the biological ammonia or ammonium oxidation to nitrite followed by two steps oxidation process carried out by nitrifying bacteria (Ali et al., 2021; Sintia et al., 2021; Nahar et al., 2020). It is an important part of the nitrogen cycle as well as considered as ecological and agricultural importance (Khanom et al., 2021; Stefanowicz et al., 2020). Antibiotic resistance genes (ARGs) are recognized as emergent environmental pollutants (Anand et al., 2021; Li et al., 2021; Rahman et al., 2018). Excessive use of pesticides in the paddy field develops ARGs and affects the nitrification activities in the paddy soils (Bashir et al., 2018; Hu et al., 2017; Ramakrishnan et al., 2019; Rahman et al., 2021; Zhao et al., 2019). The abundance of ARGs in the soil environment directly or indirectly affects the nitrification activity (Ali et al., 2021; Ma et al., 2020). When the antibiotic concentration increase, the structure of microbial populations of the soil environment changes and shows antibiotic resistance (Cycoń et al., 2019; Grenni et al., 2018; Sidhu et al., 2021). If the nitrification rate will decrease that is harmful to plants since plants indirectly take up atmospheric nitrogen by nitrification process (Moreau et al., 2019). Nevertheless, nowadays, knowledge about the response of pesticides on the nitrification activities and their interaction with heavy metal pollution, ARGs, and *amoA* genes in relation to chemical fertilizers and manure in the polluted long-term paddy

soils is very little known. For this reason, the aim of this project was i) to determine the response of pesticides on net nitrification rate (NR) and nitrification potential rate (NP), ii) to determine the heavy metal concentrations (HMCs) and quantify the abundances of some of ARGs such as *sul* genes (*sulI* and *sulII*) and *tet* genes (*tetO* and *tetQ*) and *amoA* genes (*amoA* AOA, *amoA*-AOB, and *amoA*-NOB), and iii) to find out interaction and relation among NP, NR, HMCs, ARGs and *amoA* genes in long-term treated paddy soils.

2. Materials and methods

2.1. Site specification and sample collection

The collection of soil samples was carried out from diverse paddy fields from Jashore district (23.09°N-89.09°E) in Bangladesh. Five (5) treatments, soil specimen was selected from paddy soils where cultivation times were around 8–10 (eight to ten) years such as chemical fertilizer treated paddy soil (T1), chemical fertilizer with pesticide treated paddy soil (T2), chemical fertilizer with manure treated paddy soil (T3), chemical fertilizer and manure with pesticide treated paddy soil (T4) and manure with pesticide treated paddy soil (T5), respectively. Each treatment, soil was collected from a depth of 0–20 cm and from every treatment with no less than three replicas. Each treatment, soil sample locations, number of chemical fertilizers, manure and pesticides the community farmers applied, and other details are mentioned in detail as shown in Table 1. Timing of soil sampling was from July to August 2020 during paddy cultivation. The subsequent cumulation of the soil sample, they were segregated into various portions for nitrification rate determination, pH value, and moisture content for analyzing soil chemical properties.

2.2. Analysis of physico-chemical properties of paddy soils

For chemical property analysis, test samples were partitioned into three proportions: one portion was air dried and sieved to <2 mm for physico-chemical property analysis, the second part was stored at 4 °C and processed for net nitrification analysis, and the third portion was frozen at −20 °C for molecular analysis. The physico-chemical properties, moisture content (MC), pH, total carbon (TC), soil organic carbon (SOC), total nitrogen (TN), phosphate (P), sulfur (S), potassium (K^+), calcium (Ca^{2+}) of paddy soils was determined according to Rahman et al. (2018) and Rahman et al. (2020) and Raminoarison et al. (2020).

2.3. Determination of NR and NP in different treated paddy soils

Determination of NR was terminated from the net measure of NO_3^- produced from various long-term treated paddy soils whose incubation period was 20 °C (field, limit temperature) around a 14-day time frame in accordance with Hart et al. (1997), Ali et al. (2021), Khanom et al. (2021). In sum, the overall approach strained the soil through a 4 mm sieve considering seven subsamples (App. 20 g wet soil each) of soil for each treatment and moved to 250 mL Erlenmeyer flasks with replicate. For the first subsample the inceptive NH_4^+ and NO_3^- concentration was ascertained by mixing 100 mL of 2 M KCL into the soil. For the rest six subsamples, $(NH_4)_2SO_4$ solution was mixed to dispense 75 mg N kg^{-1} and incubated at 20 °C. Further, draw out each sample at each specific period. NR was enumerated as the increase in NO_3^- concentration gradually through Nahar et al. (2020) equation:

$$NR(\text{mg N kg}^{-1} \text{ soil h}^{-1}) = \frac{(NO_3^- - N)T_2 - (NO_3^- - N)T_1}{T} \quad (1)$$

Here T is the number of hours in the middle of two sampling times (T_2 and T_1 hour), $NO_3^- - N)T_1$ and $NO_3^- - N)T_2$ are the NO_3^- concentration at that time, respectively.

Calculation of the NP was done by the slurry method (Taylor et al., 2010). In a nutshell, soil samples (10 g) were in the conical flasks (150

mL), after adding 60 mL of the phosphate solution [ammonium sulfate (1.0 mM final $(\text{NH}_4)_2\text{SO}_4$ and (1.0 mM phosphate at pH 7.2)]. Then it agitated (200 rpm) vigorously in an orbital incubator shaker at 30 °C. Subsequently 2 h, aliquot (20 mL) was sampled, and additional 20 mL was sampled after 22 h. Through a polysulfone membrane (0.45 μm), the supernatant was filtered. Before and after the experiment the soil contents of NO_3N were evaluated. Finally, the nitrification potential was enumerated in the same way as NR.

2.4. Sample digestion for heavy metal determination of paddy soils

Sample digestion for heavy metal determination of paddy soils was done according to Ali et al. (2021) and Nahar et al. (2020). In brief, soil samples were dried at 105 °C and then the soils were ground and sieved to powder by putting in a mortar with a 2 mm mesh to get a better powder. Powdered soil was processed by the strategy developed by Edori and Edori (2012) and Nahar et al. (2020). The HMCs were determined by atomic absorption spectrophotometer (A.A.S).

2.5. Determination of contamination factor (C_f) and pollution index (PI)

The C_f is the fraction of HMCs to the worth of the framework. Lacatusu's (2000) and Ali et al. (2021) technique was embraced to decide the factor's value. Then, at that point, the C_f was used to remark the phases of defilement in soil debased with HMCs. Eq. (3) addresses the C_f :

$$\text{Contamination factor } (C_f) = C_m/C_b \quad (2)$$

in which C_m is the HMC, and C_b is the value of the background which was characterized as the objective incentive for HMCs.

PI is the proportion of the level of total contamination in the sampling area. Lacatusu et al. (2000), Ali et al. (2021), and Nahar et al. (2020) technique was utilized to decide the PI for all areas, as surrendered in Eq. 4:

$$\text{PI} = (C_{f1} \times C_{f2} \times C_{f3} \times \dots \times C_{fn})^{1/n} \quad (3)$$

Where, n is the number of metals. C_f is the HMCs in the background values/heavy metals' soil. The pollution range with their significance is exposed in Table S1.

2.6. Extraction and purification of DNA from different treated paddy soils

Fresh-dried soil (0.5 g) was used to extract DNA using the FastDNA spin kit from the long-term treated paddy soils according to the manufacturer's guidelines. Purity test of DNA was applied through a

NanoDrop spectrophotometer (Wilmington, DE, USA).

2.7. Determination of antibiotic resistance genes (ARGs) and *amoA* genes (*amoA*-AOA, *amoA*-AOB, and *amoA*-NOB) by q-PCR in different treated paddy soils

ARGs (*sulI*, *sulII*, *tetO*, and *tetQ*) and *amoA* genes (*amoA*-AOA, *amoA*-AOB, and *amoA*-NOB) genes were quantified by q-PCR. For *sulI*, *sulII*, and *tetQ* genes, the primers were used as in Rahman et al. (2018) and Peng et al. (2015), similarly, primers of *amoA*-AOA and *amoA*-AOB genes were those in Khanom et al. (2021), Meinhardt et al. (2015), Rahman et al. (2021), and Sintia et al. (2021), those for *amoA*-NOB were from Ali et al. (2021), Graham et al. (2007), Khanom et al. (2021), Rahman et al. (2021), Sintia et al. (2021), respectively. The thermal condition, nucleotide sequence, primer set with other information are mentioned as shown in Table S2. For amplifying the q-PCR products, a total mixture of 20 μL SYBR green assays was applied according to Rahman et al. (2018). For data analysis and collection as well as thermal cycling, a real-time detection device was used (CFX96 Optical, Bio-Rad Laboratories Inc., Hercules, CA, USA). Standard plasmid was made from an identified DNA containing fragment genes of *sulI*, *sulII*, *tetO*, *tetQ*, *amoA*-AOA, *amoA*-AOB, and *amoA*-NOB. The standard curve was established by means of 10-fold consecutive dilutions (10^2 - 10^9 copies) of the standard plasmid. R^2 values of *sulI*, *sulII*, *tetO*, *tetQ*, *amoA*-AOA, *amoA*-AOB, and *amoA*-NOB genes were 0.99, 0.98, 0.97, 0.98, 0.99, 0.98, and 0.98, respectively. The efficiencies of amplification of *sulI*, *sulII*, *tetO*, *tetQ*, *amoA*-AOA, *amoA*-AOB, and *amoA*-NOB genes were 100, 99, 98, 101, 99, 100, and 99%, respectively. Based on the calibration curves, Ct value was calculated. Then standardized against the DNA extracted to calculate the quantify of *sulI*, *sulII*, *tetO*, *tetQ*, *amoA*-AOA, *amoA*-AOB, and *amoA*-NOB genes in per g dry soil.

2.8. Statistical analysis

A Scatter plot matrix and statistical analysis were made by SPSS and OriginPro8 software. The mean difference was assessed through ANOVA with Fisher's least significant difference (LSD) test ($p < 0.05$) was considered significant). Principal component analysis (PCA) and Pearson correlations were accomplished by using OriginPro8 software.

Table 1

Soil sample locations, soil color, coordinates, classification of soil, fertilizers used and other information in long-term treated paddy soils.

Sample no.	T1	T2	T3	T4	T5
Location	Jhikargacha, Jashore	Jhikargacha, Jashore	Jhikargacha, Jashore	Jhikargacha, Jashore	Jhikargacha, Jashore
Soil color	Gray	Gray	Dark gray	Dark gray	Gray
Coordinates	23.09°N 89.09°E	23.09°N 89.09°E	23.09°N 89.09°E	23.09°N 89.09°E	23.09°N 89.09°E
Classification of soils	Loam clay soil	Silty loam soil	Sticky clay and silty loam	Sticky clay and silty loam	Silty loam soil
Chemical fertilizers used ($\text{kg}^{-1}\text{h}^{-1}\text{y}^{-1}\text{dm}$)	TSP, 240 Urea, 320 Potash, 165	TSP, 240 Urea, 320 Potash, 165	TSP, 240 Urea, 320 Potash, 165	TSP, 240 Urea, 320 Potash, 165	No Fertilizer used
Manure ($\text{kg}^{-1}\text{h}^{-1}\text{y}^{-1}\text{dm}$)	No manure used	No manure used	Compost 35000-45000	Compost 35000-45000	Compost 35000-45000
Pesticide ($\text{kg}^{-1}\text{h}^{-1}\text{y}^{-1}\text{dm}$)	No pesticide used	Chlorantraniliprole + Thiamethoxam, 0.8	No pesticide used	Chlorantraniliprole + Thiamethoxam, 0.8	Chlorantraniliprole + Thiamethoxam, 0.8
Cultivation time (years)	8-10	8-10	8-10	8-10	8-10

Number-amounts of fertilizers used ($\text{kg}^{-1}\text{h}^{-1}\text{y}^{-1}\text{dm}$), TSP-Triple super phosphate. T1-Chemical fertilizer treated paddy soils, T2-Chemical fertilizer with pesticide treated paddy soils, T3-Chemical fertilizer with manure treated paddy soils, T4-Chemical fertilizer and manure with pesticide treated paddy soils, and T5-Manure with pesticide treated paddy soils.

3. Results

3.1. Determination of physico-chemical properties in long-term treated paddy soils

Determination of physico-chemical properties of long-term treated paddy soils, including, MC, pH, TC, TN, SOC, P, K⁺, S, Ca²⁺ was shown in Table S3. MC, pH, TC, TN, SOC, P, K⁺, S, Ca²⁺ of long-term treated paddy soils was ranged from 18.4 ± 2.3 to $19.3 \pm 2.1\%$, 7.9 ± 0.2 to 7.9 ± 0.5 , 3.2 ± 0.4 to 3.6 ± 0.7 g kg⁻¹, 2.6 ± 0.3 to 2.7 ± 0.4 g kg⁻¹, 2.9 ± 0.4 to 4.2 ± 0.4 g kg⁻¹, 27.0 ± 1.4 to 20.0 ± 2.7 g kg⁻¹, 24 ± 2.2 to 29 ± 2.1 mg g⁻¹, 49.4 ± 2.6 to 29.5 ± 2.6 g kg⁻¹, 174.5 ± 5.5 to 230.0 ± 5.8 mg g⁻¹, respectively. This finding indicated that concentrations of chemical properties of long-term treated paddy soils such as TC, SOC, TN, P, K⁺, S, and Ca²⁺ were increased significantly ($p < 0.05$) in chemical fertilizers and manure treated paddy soils. But these chemical properties were decreased significantly ($p < 0.05$) in paddy soils while treated with pesticides.

3.2. Quantification of NR and NP in long-term treated paddy soils

The NR in long-term treated paddy soils were quantified and fluctuated from 6.6 ± 0.2 to 17.1 ± 1.7 mg kg⁻¹ h⁻¹ (Fig. 1). Along with this, the NP in paddy soils was from 4.2 ± 0.5 to 14.2 ± 1.9 mg kg⁻¹ h⁻¹. The highest NR and NP were found in chemical fertilizer with manure treated paddy soils, on the other hand, the lowest was found in chemical fertilizer with pesticide treated paddy soils, demonstrating that pesticide affects the NR and NP in paddy soils. Pearson correlation analysis, among NR, NP, and chemical properties indicated that the chemical properties (pH, TC, TN, SOC, P, K, and S) were positively correlated ($p < 0.001$) with NR and NP as shown in Table S4. Scatter plot matrix analysis in long-term treated paddy soils revealed that NR and NP were linearly correlated with TC, TN, SOC, and P as shown in Fig. 2. Based on the dependent variable NR and NP, among the chemical properties, TN, TC, and SOC had linearly correlated with each other (Fig. 2).

3.3. Quantification of HMCs and their levels of contamination in long-term treated paddy soils

The quantification of HMCs in long-term treated paddy soils is presented in Table 2. The concentration of Pb²⁺, Cu²⁺, Zn²⁺, and Fe³⁺ fluctuated from 6875.6 ± 1.5 to 12536 ± 10.6 mg kg⁻¹, 28.7 ± 0.9 to 54.5 ± 2.5 mg kg⁻¹, 300.5 ± 3.9 to 456.6 ± 5.7 mg kg⁻¹, and 510.0 ± 6.7 to 820.2 ± 9.1 mg kg⁻¹, respectively. The highest concentrations of Cu²⁺, Zn²⁺, and Fe²⁺ were found in chemical fertilizer with manure treated paddy soils compared to other treated paddy soils. While the concentrations of Pb were the highest in the chemical fertilizer treated paddy soils as compared to other treated paddy soils (Table 2). According to PI analysis, moderate contamination and slight pollution for Fe³⁺ and Cu²⁺ was observed in the chemical fertilizer with manure, and chemical fertilizer and manure with pesticide treated paddy soils, respectively. Similarly, Zn²⁺ and Pb²⁺ concentrations showed the severe pollution in the manure with pesticide treated paddy soils, indicating that the application of chemical fertilizer, manure, and pesticide affects the HMCs level in soils. The Pearson correlation analysis demonstrated that NR and NP were significantly ($p < 0.001$) positively correlated with HMCs of Cu²⁺, Zn²⁺, and Fe³⁺ in the paddy soils as shown in Table S4. On the other hand, Pb²⁺ was negatively correlated with NR and NP. Scatter plot matrix analysis revealed that NR and NP were linearly correlated with Fe³⁺ in different treated paddy soils (Fig. 2C,D). The scatter plot matrix analysis also revealed that there was a linear correlation present between Fe³⁺ and Zn²⁺ among the interaction of all HMCs.

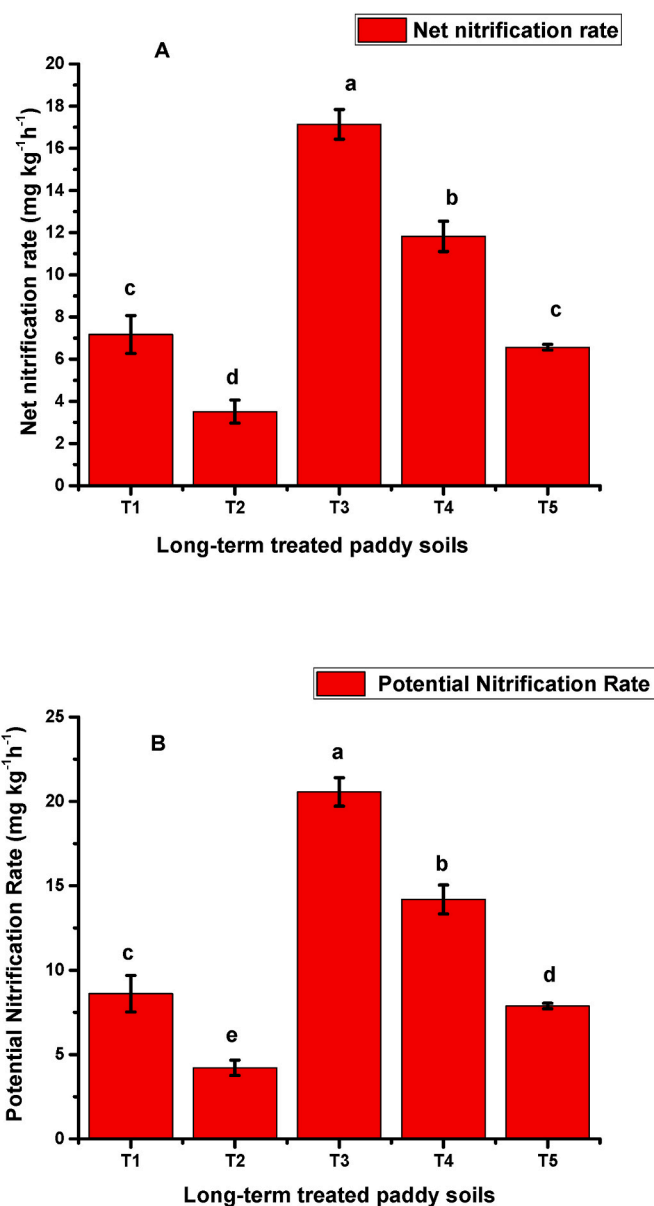


Fig. 1. NR and NP in long-term treated paddy soils. A-NR, B- NP, NR-net nitrification rate, NP- Potential nitrification rate. T1-Chemical fertilizer treated paddy soils, T2-Chemical fertilizer with pesticide treated paddy soils, T3-Chemical fertilizer with manure treated paddy soils, T4-Chemical fertilizer and manure with pesticide treated paddy soils, and T5-Manure with pesticide treated paddy soils. a, b, c, d, and e indicate significant difference at $p > 0.05$.

3.4. Abundances of ARGs and amoA genes in long-term treated paddy soils

The abundances of ARGs in long-term treated paddy soils were calculated and graphically represented as in Fig. 3A and B *sull*, *sullII*, *tetO*, and *tetQ* genes abundance were ranged from $7.1E+07 \pm 8.37E6$ to $5.09E+08 \pm 5.10E6$, $1.97E+08 \pm 7.37E+07$ to $5.08E+09 \pm 5.20E+08$, $2.35E+06 \pm 4.50E+04$ to $5.49E+07 \pm 8.91E+05$, and $3.14E+04 \pm 3.10E+02$ to $5.11E+05 \pm 4.47E+03$ per g dry soil, respectively in the long-term treated paddy soils. *amoA*-AOA, *amoA*-AOB, and *amoA*-NOB genes abundance varied from $2.20E+07 \pm 1.78E+06$ to $5.77E+8 \pm 5.07E+07$, $2.54E+06 \pm 5.15E+05$ to $4.63E+07 \pm 4.39E+06$, $2.13E+06 \pm 2.04E+05$ to $4.38E+07 \pm 8.36E+06$ per g dry soil, respectively in the long-term treated paddy soils (Fig. 3C). Pearson correlation analysis demonstrated that *sull*, *sullII*, *tetO*, and *tetQ*, *amoA*-AOA, *amoA*-AOB, and

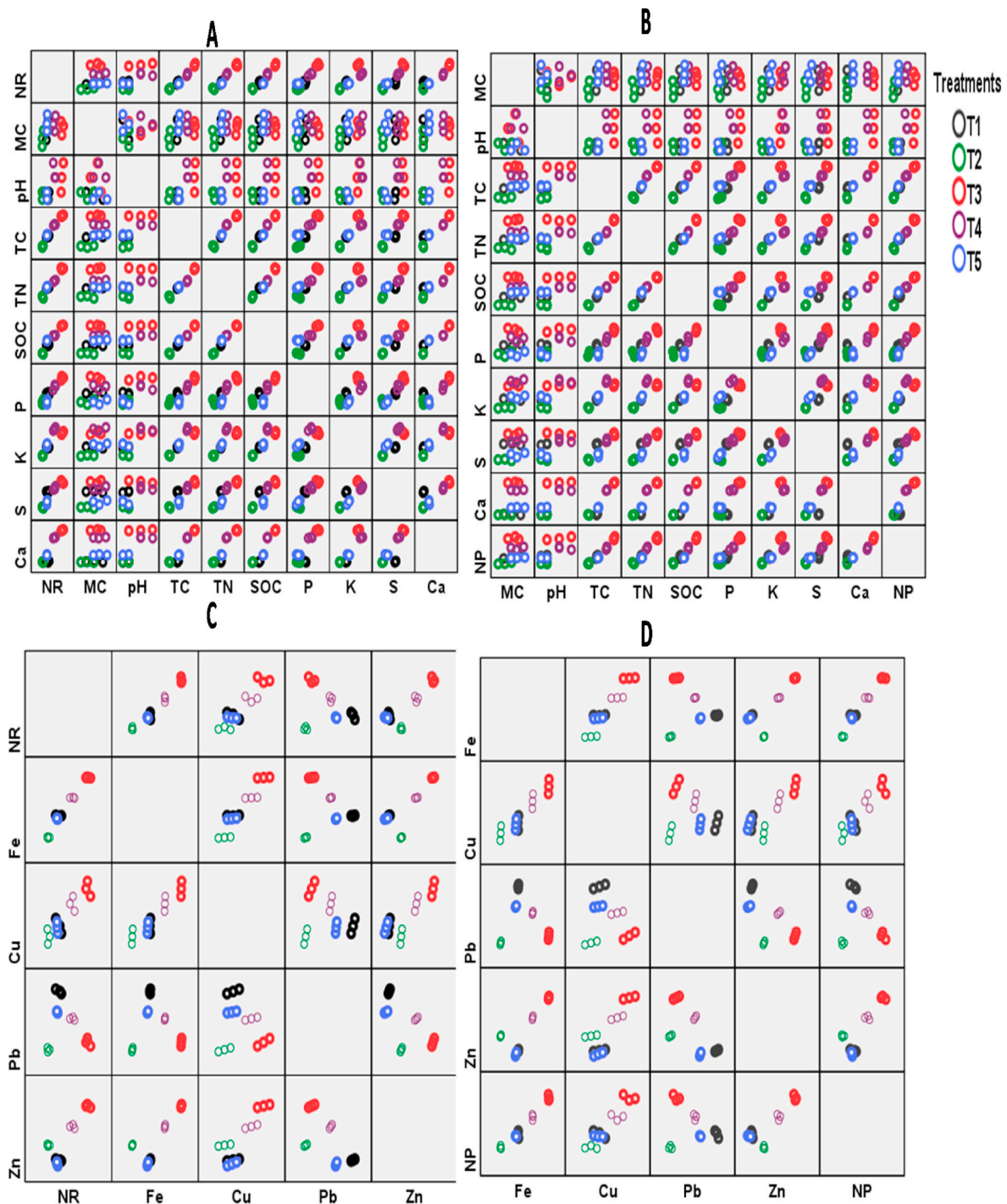


Fig. 2. Scatter plot matrix analysis in long-term treated paddy soil. A-Net nitrification rates verses chemical properties, B- Nitrification potential verses chemical properties, C-Net nitrification rates verses heavy metal concentrations, D- Nitrification potential verses heavy metal concentrations. MC- moisture content, TC- total carbon, TN- total nitrogen, SOC- soil organic carbon, NR-net nitrification rate, NP- nitrification potential. T1-Chemical fertilizer treated paddy soils, T2-Chemical fertilizer with pesticide treated paddy soils, T3-Chemical fertilizer with manure treated paddy soils, T4-Chemical fertilizer and manure with pesticide treated paddy soils, and T5-Manure with pesticide treated paddy soils.

amoA-NOB genes were significantly positive correlated ($r=0.96$ to 0.99 , $p < 0.001$ and $r=0.87$ to 0.92 , $p < 0.001$, respectively) with NR and NP (Table S5). Scatter plot matrix analysis exposed that NR and NP were linearly correlated with *sull*, *sullI*, *tetO*, and *tetQ* genes as shown in Fig. 4. On the other hand, *amoA*-AOA, *amoA*-AOB, and *amoA*-NOB were not linearly correlated with NR and NP.

3.5. Principal component analysis (PCA) in long-term treated paddy soils

PCA based on the chemical properties, HMCs, NR, NP, *sull*, *sullI*, *tetO*, *tetQ*, *amoA*-AOA, *amoA*-AOB, and *amoA*-NOB of long-term treated paddy soils was exposed in Fig. 5. The result of PCA indicated that chemical fertilizer with manure treated soils is distinct, separate from other treated paddy soils. This result also indicated that chemical fertilizer with pesticide treated soils is distinct, separate from chemical fertilizer

Table 2

HMCs and contamination level of heavy metals in long-term treated paddy soils.

Heavy metal	Treatment	Concentration (mg kg ⁻¹)	Target value (mg kg ⁻¹)	Contamination factor (CF)	Potential ecological risk
Fe ³⁺	T1	6875.6 ± 1.5d	38000	0.23	Slight contamination
	T2	8940.0 ± 10.3c		0.18	Slight contamination
	T3	12536 ± 10.6a		0.32	Moderate contamination
	T4	10637 ± 9.4b		0.27	Moderate contamination
	T5	8656.0 ± 6.5c		0.22	Slight contamination
Cu ²⁺	T1	28.7 ± 0.9d	36	0.95	Very severe contamination
	T2	34.4 ± 1.81c		0.79	Very severe contamination
	T3	54.5 ± 2.5a		1.80	Slight pollution
	T4	46.4 ± 0.9b		1.20	Slight pollution
	T5	33.6 ± 1.6c		0.93	Very severe contamination
Pb ²⁺	T1	300.5 ± 3.9e	85	5.37	Severe pollution
	T2	456.6 ± 5.7a		3.53	Moderate pollution
	T3	320.4 ± 4.6d		3.76	Moderate pollution
	T4	383.0 ± 5.2c		4.50	Severe pollution
	T5	401.0 ± 5.9b		4.71	Severe pollution
Zn ²⁺	T1	530.0 ± 7.4d	140	3.78	Moderate pollution
	T2	610.1 ± 8.6c		4.35	Severe pollution
	T3	820.2 ± 9.1a		5.85	Severe pollution
	T4	710.1 ± 8.9b		5.07	Severe pollution
	T5	510.0 ± 6.7e		3.64	Moderate pollution

a, b, c, d, and e indicate significant difference at 5% level. T1-Chemical fertilizer treated paddy soils, T2-Chemical fertilizer with pesticide treated paddy soils, T3-Chemical fertilizer with manure treated paddy soils, T4-Chemical fertilizer and manure with pesticide treated paddy soils, and T5-Manure with pesticide treated paddy soils.

treated paddy soils.

4. Discussion

4.1. Change the chemical properties of long-term treated paddy soils

Concentrations of chemical properties such as TC, TN, and SOC were significantly ($p < 0.05$) increased in chemical fertilizer with manure treated paddy soils compared to other treated paddy soils. This result indicated that chemical and manure application in paddy soils altered the soil properties and increased the concentrations of soil properties. This result is according to Rahman et al. (2018), they found that long-term manure application increase SOC, TN, and TC in paddy soils. Similarly, P, K⁺, S, and Ca²⁺ were significantly different ($p < 0.05$) in chemical fertilizer with manure treated paddy soils. This result is correlated to Sihag et al. (2005), who found that the amounts of P significantly increased in chemical fertilizers treated soils. The application of chemical fertilizer and manure with pesticide treated paddy soils significantly decreased ($p < 0.05$) the concentration of chemical properties of soil as compared with chemical fertilizer, manure, and their combination treated paddy soils, demonstrated that the application of pesticides changed the concentration of soil properties in paddy soils. This result is similar to Rahman et al. (2021). Carpio et al. (2021) reported that chemical properties of soil positively increased by the application of organic and inorganic fertilizers, while application of pesticide decreased the soil properties. Similarly, Satapute et al. (2019) reported that the application of pesticides in soil decreased the chemical properties of soil, including soil nutrient content, pH, SOC, and MC.

4.2. Effect of pesticides on NR and NP and their influencing factors

In this study, NR and NP were significantly increased ($p < 0.05$) in chemical fertilizer and manure treated paddy soils as contrasted with other treated paddy soils, exhibiting that chemical fertilizer and manure treatment enhance the nitrification activity in the paddy soils. Similarly, Nahar et al. (2020) found that chemical fertilizer and manure increased the nitrification activities in paddy soils. These outcomes additionally exhibited that manure application could increase the mineralization in soils (Wang et al., 2014). The NR and NP were significantly decreased ($p < 0.05$) in chemical fertilizer with pesticide treated paddy soils as compared with chemical treated paddy soils (Fig. 1). Similarly, the NR

and NP were significantly ($p < 0.05$) decreased in manure with pesticide treated paddy soils as compared with manure treated paddy soils. This result demonstrated that pesticide changed the concentrations of soil properties and partially inhibit nitrification activities. This result is accordance with Bei et al. (2021), they found pesticide changes the soil nitrate concentrations and nitrifying populations as well as soil nitrification activities. Correspondingly, Prudnikova et al. (2021) reported that pesticides changed the structure of soil, concentration chemical properties, and sorption properties of soil. The results of Pearson correlation indicated that NR and NP were positively correlated with chemical properties of soils, but Scatter plot matrix analysis indicated only TC, TN, SOC, and P linearly correlated. Interaction based on NR and NP among all chemical properties, only TC, TN, and SOC were linearly correlated each other's, indicating that long-term use of chemical fertilizer and manure increased the concentrations of TC, TN, and SOC in the paddy soils, which are the main factors for the nitrifying microorganisms and nitrification activities (Fig. 2). This consistent result was found from Sintia et al. (2021), they demonstrated that the TC and TN influenced the nitrification activities in cereal crop soils. Similarly, it was mentioned that TC, SOC, and TN are the main factors for the nitrifying microbes in soils (Han et al., 2021).

4.3. Effect of pesticides on HMCs in long-term treated paddy soils

In this study, Fe³⁺, Zn²⁺, Cu²⁺, and Pb²⁺ were significantly increased ($p < 0.05$) in chemical fertilizer with pesticide treated paddy soils as compared with chemical fertilizer treated paddy soils (Table 2). This result indicated that the application of pesticide increased the HMCs in soil which is indirectly and directly affected the soil properties as well as nitrification activities. This result is supported by Ali et al. (2021). Correspondingly, Nahar et al. (2021) demonstrated that pesticide may change the HMCs as well as soil properties. It is an interesting result that HMCs were significantly increased in the chemical fertilizer with manure treated paddy soils as compared with chemical fertilizer and manure with pesticide treated paddy soils, confirming the presence of inhibitory relationship due to the combinatorial interaction of manure and pesticide in the paddy soils, which may decrease the HMCs by changing the soil properties. The highest Fe³⁺ concentration (12536 ± 10.6 mg kg⁻¹) was found in chemical fertilizer with manure treated paddy soils (Table 2). This outcome was to a great extent predictable with Tariq et al. (2012) work. PI analysis revealed that Fe³⁺ was

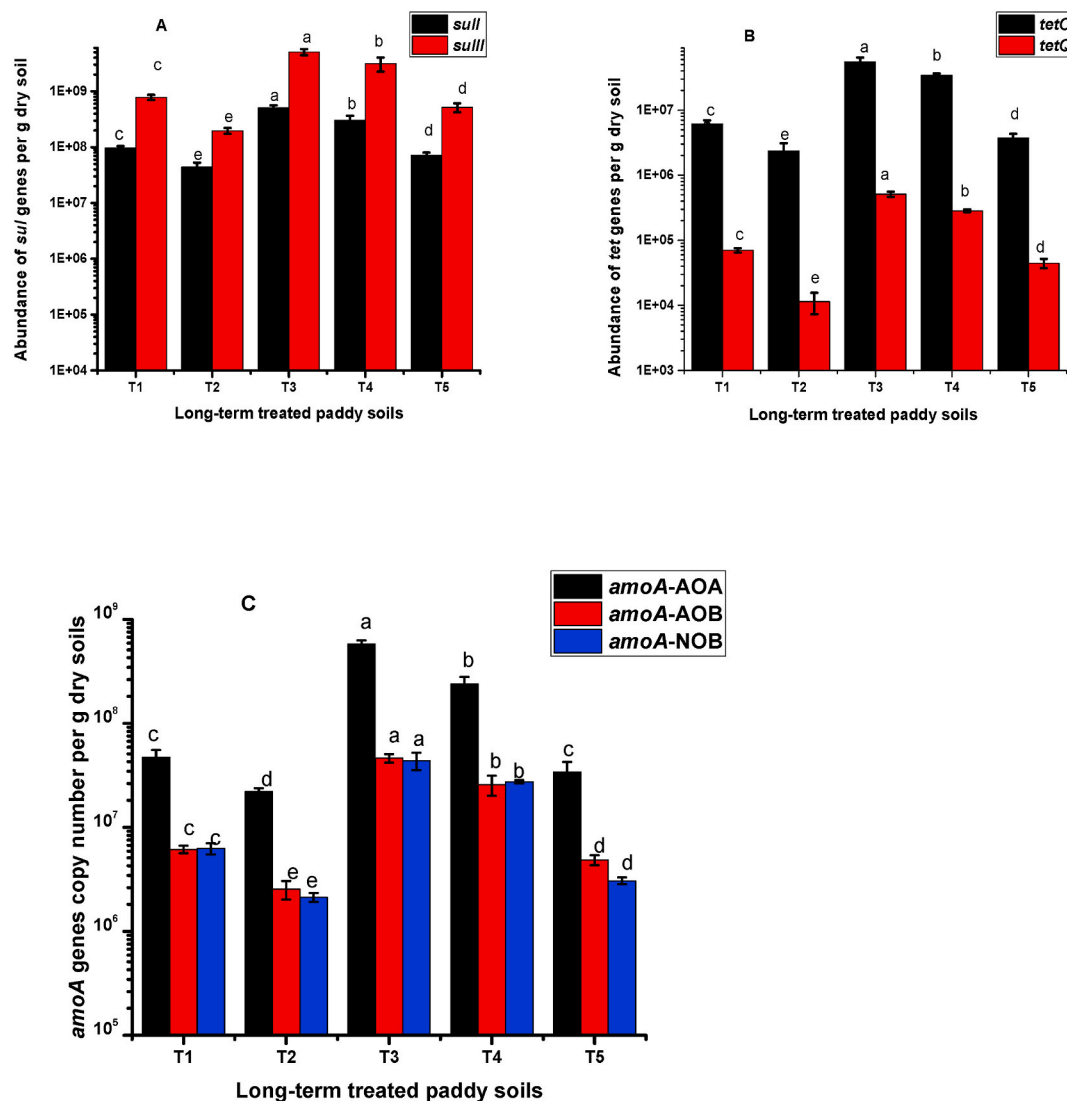


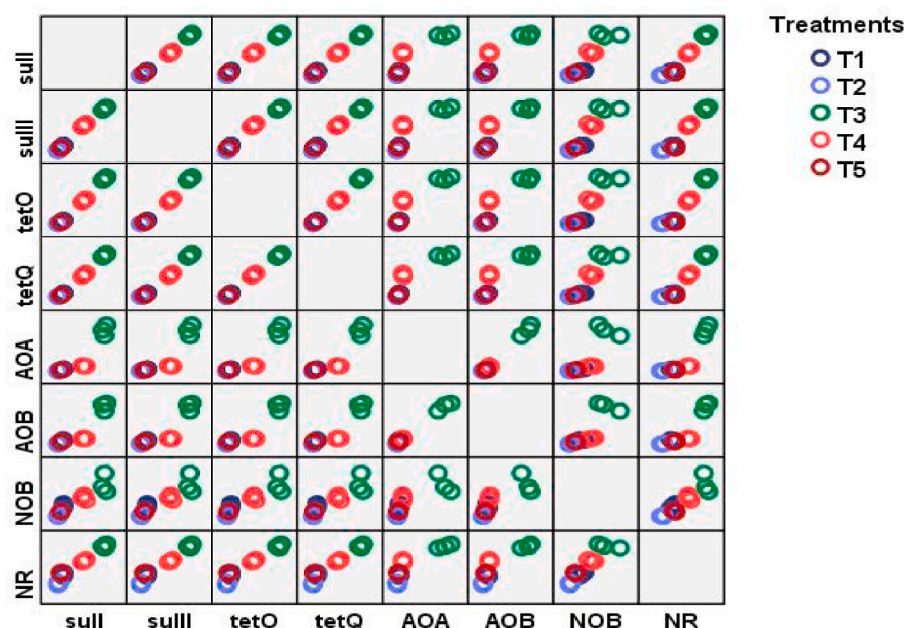
Fig. 3. Abundances of ARGs and *amoA* genes in long-term treated paddy soils. A-abundances of *sul* gene (*sulI*, *sulII*), B- abundances of *tet* gene (*tetO*, *tetQ*), and C-abundances of *amoA* gene (*amoA*-AOA, *amoA*-AOB, and *amoA*-NOB). T1-Chemical fertilizer treated paddy soils, T2-Chemical fertilizer with pesticide treated paddy soils, T3-Chemical fertilizer with manure treated paddy soils, T4-Chemical fertilizer and manure with pesticide treated paddy soils, and T5-Manure with pesticide treated paddy soils. a, b, c, d, and e indicate significant difference at $p > 0.05$.

moderately contaminated in chemical fertilizer and manure with pesticide treated paddy soils, while the others showed slight Fe contamination. It was also found that Cu^{2+} was polluted slightly in chemical fertilizer and manure with pesticide treated soils, while the rest three treated paddy soils were contaminated very severely. This result demonstrated that manure and its interaction with chemical fertilizer and pesticide are present in the paddy soils, which may influence the specific concentration of specific heavy metals. According to the Pearson correlation coefficient analysis exhibited that HMCs (except Pb^{2+}) were significantly positive correlated ($r = 0.83$ to 0.99 , $p < 0.001$) with NR and NP (Table S4), but Scatter plot matrix analysis indicated that the concentrations of Fe were linearly positive correlated with NR and NP. While the relations between NR and NP with Pb^{2+} , Cu^{2+} , and Zn^{2+} concentration were non-linearly correlated, indicating that concentrations of Fe^{2+} act as the most responsible factor, which stimulate nitrification rates and nitrification potential rates in paddy soils (Fig. 2). This result is in accordance with Huang et al. (2016), who found the most influential factor as Fe on NR in soils of China. Similarly, Dollhopf et al. (2005) found that high concentrations of Fe^{3+} stimulate nitrifying activities in paddy soils.

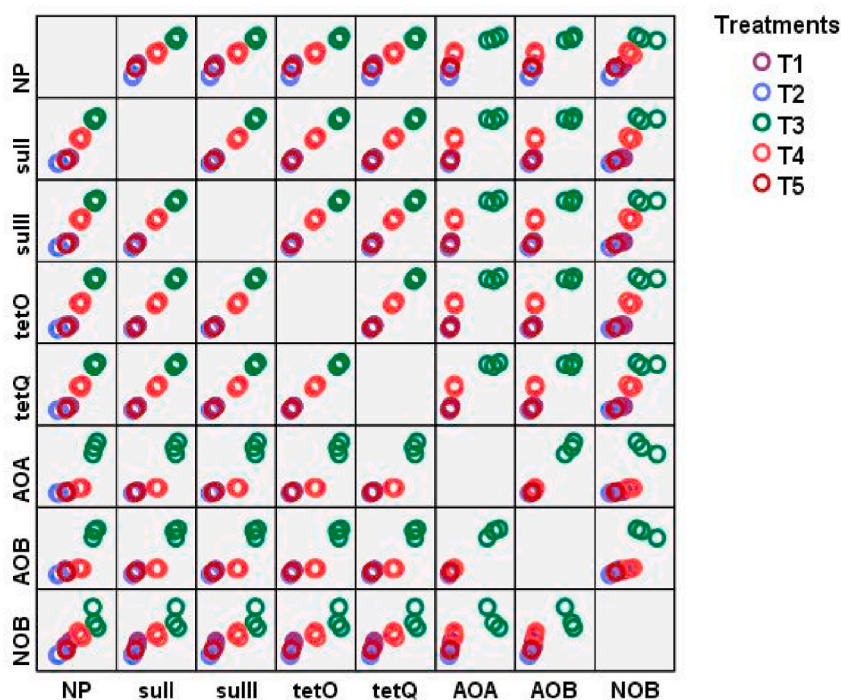
4.4. Effect of pesticides on ARGs and *amoA* genes in long-term treated paddy soils

The abundances of *sulI*, *sulII*, *tetO*, and *tetQ* genes per g of dry soils were increased in chemical fertilizer with manure treated paddy soils than any other treated paddy soils (Fig. 3A and B). There was significant variation present among chemical fertilizers with manure, manure with pesticides, and chemical fertilizer and manure with pesticide treated paddy soils. Interestingly, a significant decrease ($p < 0.05$) was found in manure with pesticide and chemical fertilizer with pesticide treated paddy soils as compared with chemical with manure treated paddy soils. These results indicated that manure increased ARGs, and pesticides decreased the abundance of ARGs in paddy soil. This result is comparable with Xu et al. (2021), who reported that the ARGs abundance are highly enriched in paddy soils by application of manure. Wang et al. (2020) and Rahman et al. (2018) noticed that manure practice noticeably improved the abundance of ARGs in soils. Similarly, Kalia et al. (2011) found that the application of pesticides had severe effects on soil ecology that decreased ARGs abundances in soils. Comparably, a higher abundances *amoA*-AOA gene found in paddy/agricultural soils as compared with that of abundances *amoA*-AOB and *amoA*-NOB genes

A



B



(Alam et al., 2013; Francis et al., 2005; Nahar et al. 2021; Jung et al., 2011). Our results showed that the abundances of all *amoA* genes were significantly increased ($p < 0.05$) in chemical fertilizer with manure treated paddy soils contrasted to other treated paddy soils. This result is indicating that chemical fertilizer and manure increased the concentration of chemical properties of soils as well as the substrate such as SOC, TN, TC, where these substrates are used as energy for ammonia oxidizers to subsidize straightforwardly the expansion in nitrifying

microbe's overflow. Consistently, it was observed that the application of N fertilizer increased the quantity of functional genes as well as nitrifying microbes in soil (Fierer et al., 2012; Liu et al., 2021; Höfferle et al., 2010). The results also exposed those abundances of *amoA* gene were significantly decreased ($p < 0.05$) in chemical fertilizer with pesticide and manure with pesticide treated paddy soils as compared with chemical fertilizer and manure treated paddy soils, demonstrated that pesticide interacts with chemical properties of soils and changes their

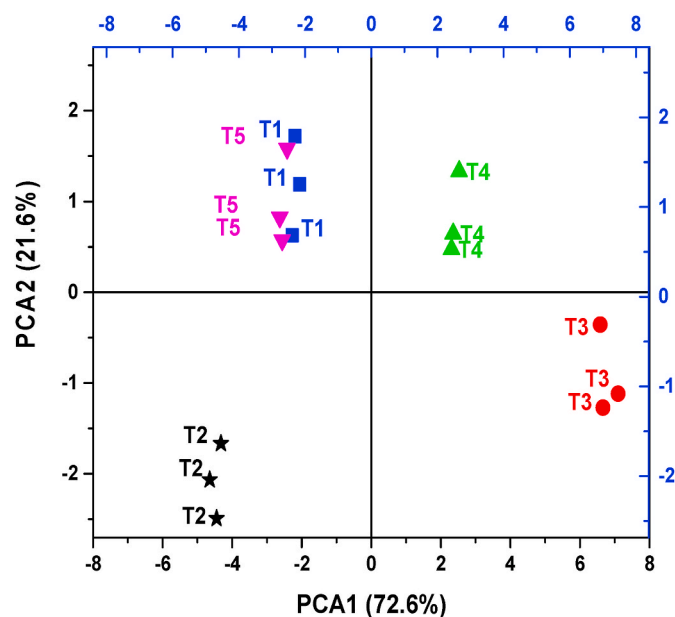


Fig. 5. PCA of different long-term paddy soils based on the chemical properties, HMCs, nitrifying activity, ARGs and *amoA* genes. T1-Chemical fertilizer treated paddy soils, T2-Chemical fertilizer with pesticide treated paddy soils, T3-Chemical fertilizer with manure treated paddy soils, T4-Chemical fertilizer and manure with pesticide treated paddy soils, and T5-Manure with pesticide treated paddy soils.

activities, which affect the growth and development of nitrifying microorganism. Similarly, previous studies indicated that nitrifying microorganism and microbial communities significantly decreased due to over used pesticide in paddy soils (Nahar et al., 2020; Rahman et al., 2020).

4.5. Relations among ARGs, *amoA* genes, NR, and NP in long-term treated paddy soils

In a previous study demonstrated that *amoA* genes positively correlated with nitrification activities in paddy soils (Yao et al., 2011; Tang et al., 2019; Rahman et al., 2021; Nahar et al., 2020). In contrast, Banning et al. (2015) reported that nitrification activities were positively correlated to *amoA*-AOB, but negatively correlated to *amoA*-AOA in acidic soils. This difference is due to long-term treated soil locations, soil types as well as practices of chemical fertilizer, pesticide, and manure. In this study, *amoA* genes and ARGs were positively correlated with NR and NP by Pearson correlation analysis (Table S5), but ARGs genes were only linearly correlated with NR and NP through Scatter plot matrix analysis (Fig. 4), demonstrated that ARGs occurred vertical gene transfer, which is responsible for nitrification activities due to the combined effect of application of manure and pesticide. Similarly, Lin et al. (2021) mentioned that ARGs proliferation mainly occurred due to vertical gene, which are involved in nitrification and ammonification. PCA also demonstrated that the effect of chemical fertilizer together with pesticide treated paddy soils completely different from the combine effect of chemical fertilizer, manure and pesticide treated paddy soils (Fig. 5), which indicates that the individual effect of pesticide as well as combine effect of pesticide with manure exist in the paddy soils.

5. Conclusions

The long-term use of chemical fertilizer, manure, and pesticide in paddy soils threatens to increase the heavy metal concentrations (HMCs) that cause human health and environmental problems. Net nitrification rate (NR), nitrification potential rate (NP), heavy metal concentrations

(HMCs), antibiotic resistance genes (ARGs), and *amoA* genes were significantly increased by the application of chemical fertilizer with manure in the long-term paddy soils. While, long-term application of pesticide in the paddy fields significantly decreased due to the NR, NP, HMCs, ARGs, and *amoA* genes. But nitrification activities are greatly affected in long-term paddy soils by total carbon (TC), soil organic carbon (SOC), total nitrogen (TN), Fe^{3+} and ARGs by Scatter plot matrix analysis. So, here combinatory effect of manure and pesticide was observed, and vertical gene transfer occurred which is responsible for nitrification activities. So, it should be paid more attention to use of pesticide and manure in paddy field soils to the possible ecological hazard of increased HMCs and ARGs.

Authors contributions

Rehena Akter, Muntaha Binte Mukhles and M. Mizanur Rahman comprehended and planned the study, carried out the analysis, wrote the manuscript; and prepared the graphs and illustrations; Md. Rasel Rana, Jannatul Ferdous, Nazmul Huda, Fahida Rahman, Meherab Hos-sain Rafi and Sudhangshu Kumar Biswas contributed to the critical revision of the manuscript and wrote the manuscript; M. Mizanur Rahman supervised the whole work, and all authors approved the final manuscript.

Ethical approve, consent to participate, and consent to publish

This article does not include any studies by any of the authors that used human or animal participants. All authors are conscious and accept responsibility for the manuscript. No part of the manuscript content has been published or accepted for publication elsewhere.

Availability of data and materials

The corresponding author M. Mizanur Rahman is responsible for all data and materials.

Ethical approval

This article does not comprise any studies with human contributors or animals performed by any of the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2022.135379>.

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